

Orals Prep

Aniruddha Madhava

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1 Overview

The following document outlines some of the notes I have taken during my oral qualifying exam preparation in the mathematics department at Stony Brook University. These notes aren't meant to be a detailed exposition, but rather a concise collection of key results and ideas that I should remember. In some places, I may add a bit of exposition to emphasize a particular idea or result.

This document is split into two main parts: (1) the major topic, which is (hyperbolic) partial differential equations, and (2) the minor topic, which is fluids (tbd). Within the major topic part, I split the document further into smaller classifications.

As always, the aim is to reduce the errors as much as possible. However, it is inevitable that some type of error (whether a typo, or something more substantial) makes it way into this document. I take full responsibility for these errors.

Part I

Hyperbolic Partial Differential Equations